

Prerequisites for setting up EC2 for VTON

These are the activities which Dinesh followed before handing over the EC2 to the AI team to proceed with the vton model inference script deployment. To carry out these steps, ensure your IAM user has access to the following:

- EC2
- VPC
- Service Quotas
- IAM

Also, ensure that the G5 instance is available in the region where you plan to set up. The major steps are listed below:

1. Configure Network

VPC and Subnets:

- Go to **VPC > Create VPC**.
- Select **VPC and more** option, which will enable us to create VPC, Subnets, NAT gateways, Route Tables at once
- Set CIDR block for the VPC (example: `10.0.0.0/16`).
- Set CIDR for public and private subnets.
- NAT gateway
- Enable DNS hostnames & DNS resolution(recommended).

The screenshot displays the 'Create VPC' wizard in the AWS Management Console. The left sidebar contains the 'VPC settings' panel, and the main area shows a 'Preview' of the network configuration.

VPC settings:

- Resources to create:** ☒ VPC and more
- Name tag auto-generation:** ☒ Auto-generate (Name tag: project)
- IPv4 CIDR block:** 10.0.0.0/16 (CIDR block size must be between /16 and /28)
- IPv6 CIDR block:** ☒ No IPv6 CIDR block
- Tenancy:** Default
- Number of Availability Zones (AZs):** 2
- Number of public subnets:** 2

Preview:

- VPC:** project-vpc
- Subnets (4):** project-subnet-public-1-ap, project-subnet-public-2-ap, project-subnet-private-1-ap, project-subnet-private-2-ap
- Route tables (3):** project-rtb-public, project-rtb-private-1-ap-southeast-2a, project-rtb-private-2-ap-southeast-2b
- Network connections (2):** project-igmp, project-igmp-s3

VPC > Your VPCs > Create VPC

0 | **2**

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | **2** | 4

▼ **Customize subnets CIDR blocks**

Public subnet CIDR block in ap-southeast-2a
10.0.0.0/20 4,096 IPs

Public subnet CIDR block in ap-southeast-2b
10.0.16.0/20 4,096 IPs

Private subnet CIDR block in ap-southeast-2a
10.0.128.0/20 4,096 IPs

Private subnet CIDR block in ap-southeast-2b
10.0.144.0/20 4,096 IPs

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.
None | **In 1 AZ** | 1 per AZ

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.
None | **S3 Gateway**

DNS options [Info](#)
☒ Enable DNS hostnames
☒ Enable DNS resolution

► **Additional tags**

[Cancel](#) [Preview code](#) [Create VPC](#)

2. Create IAM Role

- Navigate to IAM > Roles > Create Roles
- Select 'AWS service'
- Use case: EC2
- Click on next.

IAM > Roles > Create role

Step 1 **Select trusted entity** [Info](#)
Step 2 Add permissions
Step 3 Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.
Use case
☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

- Search for below-mentioned policies and select

- [CloudWatchAgentServerPolicy](#)
 - [AmazonSSMManagedInstanceCore](#)
 - [AmazonSQSFullAccess](#)
 - [AmazonKendraFullAccess](#)
 - [AmazonDynamoDBFullAccess](#)
 - [AmazonBedrockFullAccess](#)
 - [AmazonS3FullAccess](#)
- Click on next after selecting all the mentioned policies.

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

Add permissions [info](#)

Choose one or more policies to attach to your new role.

Filter by Type

All types

1 match

☒	Policy name ?	▲ Type
☒	☐ AmazonSSMManagedInstanceCore	AWS managed

► Set permissions boundary - optional

- Give a name for the Role and create the role.

IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.

Maximum 64 characters. Use alphanumeric and "+, -, @, _" characters.

Description
Add a short explanation for this role.

Maximum 1000 characters. Use letters [A-Z and a-z], numbers [0-9], tabs, new lines, or any of the following characters: "_", "-", "@", "/", "[", "]", "%", "&", "=", ":", ";".

Trust policy

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "sts:AssumeRole"  
8       ],  
9       "Principal": {  
10        "Service": [  
11          "ec2.amazonaws.com"  
12        ]  
13      }  
14    ]  
15  }  
16 }
```

Step 2: Add permissions

3. Service Quota Check and Increase

- Navigate to **Service Quotas > AWS Services > EC2**.
- Search for **Running On-Demand G and VT instances**.
- Submit a request for **g5.2xlarge** based on expected usage.

The screenshot shows the Amazon Elastic Compute Cloud (Amazon EC2) console. The left sidebar contains navigation links for 'Service Quotas', 'AWS services', 'Amazon Elastic Compute Cloud (Amazon EC2)', 'Service Quotas', 'Backward', 'AWS services', 'Quota request History', 'Organization', and 'Quota request template'. The main content area is titled 'Amazon Elastic Compute Cloud (Amazon EC2)' and 'Service quotas'. It includes a search bar with the text 'Running On-Demand G and VT instances' and a button '1 match'. Below the search bar is a table with the following columns: 'Quota name', 'Applied account-level quota value', 'AWS default quota value', 'Utilization', and 'Adjustability'. The table contains one row with the following data: 'Running On-Demand G and VT instances', '16', '0', '0%', and 'Adjustability'.

Quota name	Applied account-level quota value	AWS default quota value	Utilization	Adjustability
Running On-Demand G and VT instances	16	0	0%	Adjustability

- To request, click on the “Request increase at account level”.

Note: Since g5.2xlarge instance has 8 vCPU, quota value should be 8vCPU for 1 instance, 16 vCPU for 2 instances and so on.

Request quota increase: Running On-Demand G and VT instances

Description
Maximum number of vCPUs assigned to the Running On-Demand G and VT instances.

Requested for
Account (730335611421)
Region
Asia Pacific (Sydney) ap-southeast-2

Increase quota value
Enter in the total amount that you want the quota to be.

Must be a number greater than your current quota value of 16

Utilization
8

Approvals: For some services, smaller increases are automatically approved, while larger requests are submitted to AWS Support.

Approval timeline: AWS Support can approve, deny, or partially approve your requests. Larger increase requests take more time to process and assess while we work with the service team. For urgent requests, use [AWS Support Center](#).

[Cancel](#) [Request](#)

- Once requested, the quota increase is not immediate and will take some time for AWS to review and increase. A case will be created. You can check AWS Support > Your Support Cases.

You can check the status or follow up with AWS for the latest update.

AWS Support

>

Your support cases

>

Details

Support Center

<

Account number: 730335611421

Support plan: Basic [Change](#)

[View support plans](#)

Your support cases

AWS Support App in Slack

[Compute Optimizer](#)

[AWS Health Dashboard](#)

[Trusted Advisor](#)

▼ AWS Knowledge resources

[Knowledge Center](#)

[Knowledge Center videos](#)

[AWS Documentation](#)

[re:Post](#)

[Training and Certification](#)

Case ID 173027221300519

Info

Case details

Subject
Limit Increase: EC2 Instances

Case ID
173027221300519

Created
2024-10-30T07:10:13.455Z

Case type
Service limits

Opened by
cloudr00t@groome.ai

Status
Resolved

Severity
General question

Category
Service Limit Increase, EC2 Instances

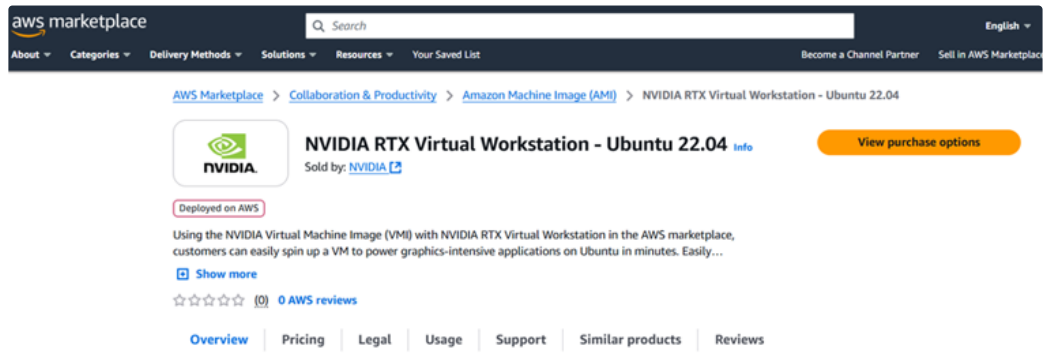
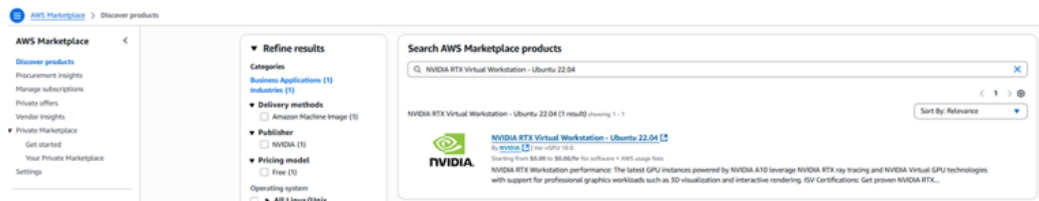
Language
English

Additional contacts
-

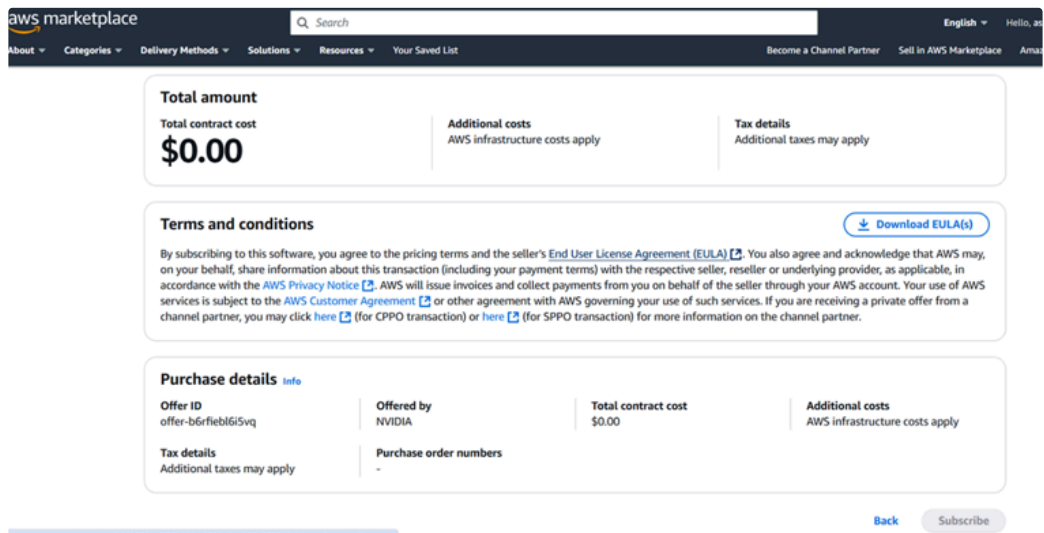
Correspondence

4. AMI Subscription from AWS Marketplace

- Navigate to AWS Marketplace service > Discover Products
- Search for NVIDIA RTX Virtual Workstation - Ubuntu 22.04 and click on the product.
- This AMI is provided by Nvidia with no subscription cost.

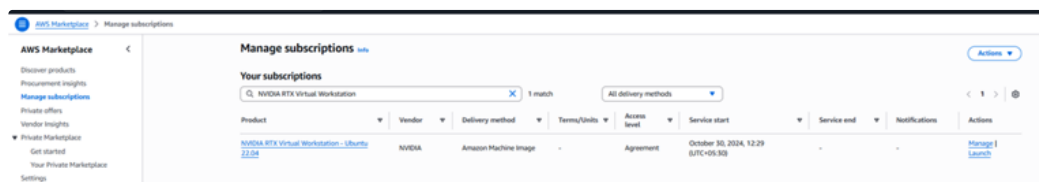


- Click on View purchase options. You can see the subscribe button in the bottom.

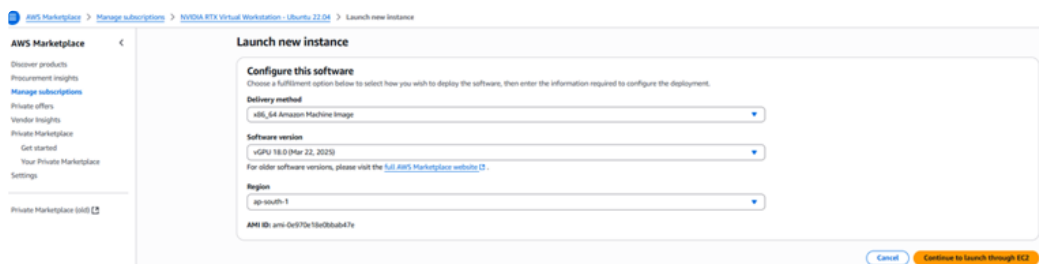


5. Launching the EC2 instance

- Once subscribed, you can view it in AWS Marketplace > Manage Subscriptions



- Click on Launch → Continue to launch through EC2



- Select Instance type: g5.2xlarge

- ▼ Instance type [Info](#) | [Get advice](#)

Instance type

g5.2xlarge

Family: g5 8 vCPU 32 GiB Memory Current generation: true

All generations

Compare instance types

The AMI vendor recommends using a g5.16xlarge instance (or larger) for the best experience with this product.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

dev

Create new key pair

- ▼ Network settings

Info

VPC - required | Info

vpc-0a9edb09c4b93c38a (np-vpc-01)

10.0.0.0/16

Subnet | Info

subnet-0c3326c46c21651b2

np-priv1-ap-southeast-2a

VPC: vpc-0a9edb09c4b93c38a Owner: 730335611421 Availability Zone: ap-southeast-2a
Zone type: Availability Zone IP addresses available: 1012 CIDR: 10.0.0.0/22

Create new subnet

Auto-assign public IP | Info

Disable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

NVIDIA RTX Virtual Workstation - Ubuntu 22.04-vGPU 18.0-AutogenByAWSMP--1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./!@#%&'()*~=:;`{|}~\$*

Description - required | Info

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Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 10.0.0.0/16)

Type | Info

ssh

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Custom

Source | Info

Q [Add CIDR, prefix list or security group]

10.0.0.0/16 X

Description - optional | Info

e.g. SSH for admin desktop

Remove

- **Security Group:**
 - Select create security group.
 - Inbound Security Group rules:
 - Type: ssh
 - Source Type: Custom
 - Source: Select VPC CIDR range (10.0.0.0/16)
- Select 100GB for EBS storage and leave the remaining settings as default.

- Verify the summary on the right and click on Launch instance.

6. Post-Launch Setup

Add the IAM role that was created previously to the launched instance.

[illegible]